Module 2 — Read 1

Binomial center and spread through confidence intervals and sampling

Module 2, Read 1: Estimation and sampling

Project: Project 2 — Estimating the Truth · **Preview:** Read 0 · **Prerequisite:** Module 1 · **Next:** Quiz A (Def & Notation), then Read 2.

This reading is your **self-paced textbook** for Module 2, Part 1. Work one **section** at a time; use **Check yourself** boxes before Quiz A. You will estimate a **population proportion or mean** with a **95% confidence interval** from an **SRS**.

Learning outcomes

Work through all five sections below in order. Each section ends with **Check yourself** questions.

After this reading you should be able to:

Section 1: Binomial mean and standard deviation

Mean and standard deviation of a binomial random variable

- Use $\mu = np$ and $\sigma = \sqrt{np(1-p)}$ for $X \sim \text{Binomial}(n, p)$.
- Interpret the mean as the **center** and the SD as **typical deviation** of X.
- Identify n and p in applied settings and compute μ and σ.

Section 2: Normal approximation and the Empirical Rule

Normal approximation to the binomial; one-proportion test by hand

- Use the **CLT** to approximate a binomial distribution by a normal curve with the same μ and σ.
- Check conditions such as $np \geq 10$ and $n(1-p) \geq 10$ when appropriate.
- State the **Empirical Rule** for normal distributions.
- Sketch a **rejection region** under a normal curve for a one-proportion test and relate the shaded area to the p-value (approximate).

Section 3: Sampling distributions and confidence intervals

Sampling distributions and estimation

- Define the **sampling distribution** of a statistic.
- Describe **confidence level**, **parameter**, and the endpoints of a confidence interval in context.
- Compute and interpret a confidence interval for a **population proportion** and for a **population mean** (methods from lecture).

Section 4: Confidence intervals and the importance of sampling

Confidence intervals and the importance of sampling

- Discuss **bias** and its impact given how a sample was collected.
- Identify ways to reduce bias and increase **precision**.
- Identify a **sampling frame** and carry out **simple random sampling**.
- Classify methods as **convenience**, **SRS**, **cluster**, **systematic**, **stratified**, or **multi-stage**.

Section 5: Sampling methods in practice

Sampling methods in practice

- Identify a **sampling frame** for a stated population.
- **Carry out** simple random sampling using a numbered list and random draws.
- **Classify** sampling designs and discuss bias, generalizability, and how to improve the design.

Section 1: Binomial mean and standard deviation

Binomial mean and standard deviation

> Bridge from Module 1
>
> You already know the binomial pmf from Module 1 Read 1. Here we summarize the **center** ($\mu = np$) and **spread** ($\sigma = \sqrt{np(1-p)}$) of the count X — the same μ and σ used when we approximate a binomial by a normal curve.

For a binomial random variable X with n trials and success probability p ($X \sim \text{Binomial}(n, p)$):

Mean (expected value):

$$\mu = np$$

Standard deviation:

$$\sigma = \sqrt{n\,p\,(1-p)}$$

Interpretation

- **Mean μ:** The **center** of the distribution of X. If we observed X many, many times (each time from a new binomial process with the same n and p), the **average** of those values would be $\mu = np$.
- **Standard deviation σ:** A **typical deviation** of X from the center. It is the square root of the average squared distance of (many) observations of X from the mean μ. So σ answers: "Roughly how far from μ do we expect a single observation of X to fall?"

Examples: Identify n and p, then compute μ and σ

For each scenario, (1) state what X counts and verify the setting is a binomial random process (binary outcome, fixed n, fixed p, independent trials); (2) identify n and p; (3) compute the mean $\mu = np$ and the standard deviation $\sigma = \sqrt{np(1-p)}$.

Example 1: ESP test

We run an ESP test with **25 trials**. On each trial the person guesses one of **5 symbols**; the computer picks one at random. So by chance the probability of a correct guess on any one trial is $p = 1/5 = 0.2$. Let X = number of correct guesses out of 25.

- X **counts:** number of correct guesses in 25 trials.
- **Binomial?** Binary (correct or not), fixed $n = 25$, fixed $p = 0.2$, independent trials. Yes.
- n **and** p**:** $n = 25$, $p = 0.2$.
- **Mean:** $\mu = np = 25 \times 0.2 = 5$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{25 \times 0.2 \times 0.8} = \sqrt{4} = 2$.

So we expect about 5 correct by chance, with a typical deviation of about 2.

Example 2: Left-handed students

We take a random sample of **30 students** from our school. About **10%** of people are left-handed nationwide; we use $p = 0.10$ as the probability that a randomly chosen student is left-handed. Let X = number of left-handed students in the sample.

- X **counts:** number of left-handed students in 30 students.
- **Binomial?** Binary (left-handed or not), fixed $n = 30$, fixed $p = 0.10$, independent students. Yes.
- n **and** p**:** $n = 30$, $p = 0.10$.
- **Mean:** $\mu = np = 30 \times 0.10 = 3$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{30 \times 0.10 \times 0.90} = \sqrt{2.7} \approx 1.64$.

We expect about 3 left-handed students, with a typical deviation of about 1.64.

Example 3: Defective items

A supplier claims that **12%** of items are defective. We inspect **20 items** at random. Let X = number of defective items among the 20.

- X **counts:** number of defective items in 20 inspected.
- **Binomial?** Binary (defective or not), fixed $n = 20$, fixed $p = 0.12$ (under the supplier's claim), independent items. Yes.
- n **and** p**:** $n = 20$, $p = 0.12$.
- **Mean:** $\mu = np = 20 \times 0.12 = 2.4$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.12 \times 0.88} = \sqrt{2.112} \approx 1.45$.

We expect about 2.4 defectives, with a typical deviation of about 1.45.

Example 4: Free throws

A basketball player shoots **40 free throws**. Her long-run probability of making any single free throw is **0.75**. Let X = number of made free throws out of 40.

- X **counts:** number of made free throws in 40 attempts.
- **Binomial?** Binary (make or miss), fixed $n = 40$, fixed $p = 0.75$, independent shots. Yes.
- n **and** p**:** $n = 40$, $p = 0.75$.
- **Mean:** $\mu = np = 40 \times 0.75 = 30$.
- **SD:** $\sigma = \sqrt{np(1-p)} = \sqrt{40 \times 0.75 \times 0.25} = \sqrt{7.5} \approx 2.74$.

We expect about 30 makes, with a typical deviation of about 2.74.

Summary

Example	X counts	n	p	$\mu = np$	$\sigma = \sqrt{np(1-p)}$
ESP	correct guesses	25	0.20	5	2
Left-handed	left-handed in sample	30	0.10	3	≈ 1.64
Defective	defectives in 20 items	20	0.12	2.4	≈ 1.45
Free throws	made shots in 40	40	0.75	30	≈ 2.74

In each case, μ is the center of the distribution (the average of many observations of X), and σ is a typical deviation from that center (square root of the average squared distance from μ).

Check yourself — Binomial mean and standard deviation

1. ESP test: 25 trials, $p = 0.2$. What are μ and σ for X = number correct?
2. 30 students, $p = 0.10$ left-handed. Compute μ and σ for the count of left-handers.
3. In words, what does $\mu = np$ represent for a binomial count X?
4. Free throws: $n = 40$, $p = 0.75$. Compute μ and σ.

Answers — Binomial mean and standard deviation

1. $\mu = np = 25(0.2) = 5$. $\sigma = \sqrt{np(1-p)} = \sqrt{25(0.2)(0.8)} = 2$.
2. $\mu = 3$. $\sigma = \sqrt{30(0.10)(0.90)} \approx 1.64$.
3. The **center** — the long-run average number of successes if the process were repeated many times.
4. $\mu = 30$. $\sigma = \sqrt{40(0.75)(0.25)} \approx 2.74$.

Section 2: Normal approximation and the Empirical Rule

In this section

- **Use the CLT to approximate the Binomial Distribution with the Normal Distribution.**
- **State the Empirical Rule for the normal distribution.**
- **Draw a rejection region by hand to test a null hypothesis for one proportion/probability p.**

Normal approximation to the binomial (CLT)

When $X \sim \text{Binomial}(n, p)$ with n large enough, the distribution of X is approximately **normal** with the same mean and standard deviation:

- **Mean:** $\mu = np$
- **Standard deviation:** $\sigma = \sqrt{n\,p\,(1-p)}$

So we can approximate $P(X \le x)$ or $P(X \ge x)$ using a normal curve with that μ and σ.

Conditions for the approximation to be reasonable: Both the expected number of successes and the expected number of failures under the null are at least 10:

- $n\,p \ge 10$
- $n\,(1-p) \ge 10$

This keeps the binomial from being too skewed so the normal curve fits reasonably well.

Empirical Rule (normal distribution)

For a **normal distribution** with mean μ and standard deviation σ:

- About **68%** of the distribution lies within **1** standard deviation of the mean: between $\mu - \sigma$ and $\mu + \sigma$.
- About **95%** lies within **2** standard deviations: between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About **99.7%** lies within **3** standard deviations: between $\mu - 3\sigma$ and $\mu + 3\sigma$.

Visualize with the One Proportion applet

Use the **One Proportion** applet to see the binomial distribution and its normal approximation

One Proportion applet (optional — compare binomial bars to the normal overlay)

Try different values of n and p: when $np \ge 10$ and $n(1-p) \ge 10$, the normal curve overlays the binomial bars closely. When those conditions fail, the binomial is skewed and the normal fit is poor.

Drawing a rejection region by hand

To test a null hypothesis about a proportion (e.g. $H_0 : p = 0.5$ for a fair coin):

1. Under H_0, identify n and p; compute $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.
2. Sketch a **normal curve** centered at μ with spread σ (or mark μ, $\mu \pm \sigma$, $\mu \pm 2\sigma$ using the Empirical Rule).
3. Mark the **observed value** of x (the count) on the horizontal axis.
4. **Shade the rejection region**: the tail(s) in the direction of H_a (e.g. for $H_a : p > 0.5$, shade the right tail; for $H_a : p \neq 0.5$, shade both tails).
5. The **p-value** is the area of the shaded region under the normal curve (approximate). If that area is small, reject H_0; if it is large, fail to reject H_0.

Example 1: Testing a coin (reject H_0)

Research question: Is the coin fair? We test $H_0 : p = 0.5$ vs $H_a : p \neq 0.5$, where $p =$ probability of heads on a single flip.

Data: We flip the coin **100** times and observe **35** heads.

- n and p under H_0: $n = 100$, $p = 0.5$.
- **Check conditions:** $np = 50 \geq 10$ and $n(1-p) = 50 \geq 10$. OK to use the normal approximation.
- **Mean and SD:** $\mu = 100 \times 0.5 = 50$, $\sigma = \sqrt{100 \times 0.5 \times 0.5} = 5$.
- **Observed:** $x = 35$ heads. Under a normal curve with mean 50 and SD 5, 35 is **3 standard deviations** below the mean ($z = (35-50)/5 = -3$). The Empirical Rule says only about 0.15% of the distribution is below $\mu - 3\sigma$, so the **left tail** (and the symmetric right tail for a two-sided test) is very small.
- **Rejection region:** Shade both tails (since $H_a : p \neq 0.5$). The observed 35 falls far into the left tail.
- **p-value:** The area in the left tail (and the symmetric right tail) is very small. So the **p-value is small**.
- **Conclusion: Reject** H_0. The data are inconsistent with a fair coin; we have evidence that $p \neq 0.5$ (here, fewer heads than expected).

Example 2: Testing a die (fail to reject H_0)

Research question: Is the die fair for the face "6"? We test $H_0 : p = 1/6$ vs $H_a : p \neq 1/6$, where $p =$ probability of a 6 on a single roll.

Data: We roll the die **60** times and observe **12** sixes.

- n and p under H_0: $n = 60$, $p = 1/6$.
- **Check conditions:** $np = 60 \times (1/6) = 10 \geq 10$ and $n(1-p) = 60 \times (5/6) = 50 \geq 10$. OK to use the normal approximation.
- **Mean and SD:** $\mu = 60 \times (1/6) = 10$, $\sigma = \sqrt{60 \times (1/6) \times (5/6)} = \sqrt{50/6} \approx 2.89$.
- **Observed:** $x = 12$ sixes. Under a normal curve with mean 10 and SD 2.89, $z = (12-10)/2.89 \approx 0.69$. So 12 is a bit **above** the mean, within about one standard deviation. Most of the distribution lies between $\mu - 2\sigma$ and $\mu + 2\sigma$ (about 95%); 12 is well inside that range.
- **Rejection region:** For $H_a : p \neq 1/6$, we would shade both tails. The observed 12 is **not** in the tails; it's near the center.
- **p-value:** The area in the tails (values as or more extreme than 12) is **large** — 12 is a typical outcome when $p = 1/6$.
- **Conclusion: Fail to reject** H_0. The data are consistent with a fair die; we do not have evidence that $p \neq 1/6$.

Summary

Example	H_0	n	Observed x	μ	σ	Location of x	p-value	Conclusion
Coin	$p = 0.5$	100	35 heads	50	5	Far in left tail	Small	Reject H_0
Die	$p = 1/6$	60	12 sixes	10	≈ 2.89	Near center	Large	Fail to reject H_0

Use the One Proportion applet to enter these n and p, see the normal overlay, and confirm the rejection regions and conclusions. ::: {.callout-tip icon=false} ## Check yourself — Normal approximation and the Empirical Rule

1. When is the normal approximation to a binomial reasonable? State both conditions.
2. State the **Empirical Rule** for a normal distribution.
3. Fair coin, $n = 100$, $H_0 : p = 0.5$, observe $x = 35$. Under H_0, what are μ and σ?
4. For a two-sided test, where do you shade the rejection region on a normal sketch?

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Answers — Normal approximation and the Empirical Rule

1. $np \geq 10$ and $n(1-p) \geq 10$ (expected successes and failures both at least 10).
2. About 68% within 1 SD of μ, 95% within 2 SD, 99.7% within 3 SD.
3. $\mu = 50$, $\sigma = 5$. The value 35 is 3 SD below the mean ($z = -3$).
4. Both tails — outcomes as or more extreme in either direction.

Section 3: Sampling distributions and confidence intervals

In this section

- Define the "sampling distribution" of a statistic.
- Describe confidence level, parameter, and upper and lower bounds of an interval in the context of an estimation problem.
- Compute a confidence interval for a population proportion.
- Interpret a confidence interval for a population proportion.
- Compute a confidence interval for a population mean.
- Interpret a confidence interval for a population mean.

Why estimation?

Descriptive statistics summarize the information in a dataset. The goal of **inferential statistics** is to infer something about a **population** from a **sample**.

Definitions: - **Population vs sample:** We observe information from a subset of people or objects: this set is the **sample**; in inferential statistics, we want to say something about the population (e.g. all people or objects in our research question), not just the ones in our sample. - **Parameter vs statistic:** A **parameter** is a number describing the population: usually this is unknown as we only have information from a subset of the population (a sample); a **statistic** is a number computed from the sample (e.g. sample proportion, sample mean).

Example: Is Moderna's vaccine effective?
Moderna vaccine trial (Reuters)

What is the population in this context? The sample? What is the parameter and what is the statistic?

Worked solution — vaccine trial

- **Population:** all people who could receive the vaccine.
- **Sample:** people actually enrolled and measured.
- **Parameter:** true efficacy / infection rate in the population.
- **Statistic:** observed rate in the sample.

Example setting: St. Joe's in Eureka, California

St. Joe's Hospital in Eureka, California performs a certain **heart procedure**. We are interested in whether the observed mortality rate at St. Joe's for this procedure is higher than the national rate.

National mortality rate (benchmark): $p_0 = 0.08$ (8%).

Two sets of data:

- **Large sample:** 200 patients, with 14 deaths ($x = 14$, $n = 200$).

- **Small sample:** 16 patients, with 3 deaths ($x = 3$, $n = 16$).
- **Population:** All patients who could receive this heart procedure (or all such patients at similar hospitals).
- **Parameter:** The true mortality rate (proportion) at St. Joe's, often denoted π or p.
- **Statistic:** The proportion who died in the sample: $\hat{p} = x/n$.

General notation and sampling distribution (proportion)

For a binary outcome (e.g. died vs. survived):

- **Parameter:** population proportion π (or p).
- **Statistic:** sample proportion $\hat{p}$ = (number of successes) / n.

The **sampling distribution** of $\hat{p}$ can be:

- Simulated (e.g. using a for loop and mean function in R with a large number of replicates),
- Computed exactly using the binomial distribution (e.g. dbinom/pbinom in R),
- Approximated with the normal distribution when n is large enough (e.g. np and n(1-p) are both at least 10).

Use the One Proportion applet to compare the observed sample proportion to the sampling distribution if St. Joe's had the same mortality rate as the national rate (8%).

General formula for a 95% confidence interval

For a **95% confidence interval**, we use the "2 standard deviations" rule from the Empirical Rule (approximate):

Statistic ± 2 × (standard deviation of statistic)

The "standard deviation of the statistic" is also called the **standard error (SE)**. So:

95% CI statistic ± 2 × SE

> Definition: 95% confidence level
>
> If we repeated sampling many times and built a 95% CI each time, about **95%** of those intervals would contain the true parameter.

Two important cases:

Formula 1: One proportion (Wald formula)

For a **population proportion** π, we estimate it with the sample proportion $\hat{p}$. The standard error of $\hat{p}$ is:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

So the **95% confidence interval for a single proportion** is:

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Conditions for Formula 1 (Wald):

1. **Binomial random process:** (a) **B**inary outcome (success/failure), (b) **I**ndependent trials, (c) **N**umber of trials n fixed in advance, (d) **S**ame probability of success π on each trial.

2. **Normal approximation:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ (at least 10 successes and 10 failures in the sample, so the sampling distribution of $\hat{p}$ is approximately normal).

Formula 2: One mean

For a **population mean** μ, we estimate it with the sample mean $\bar{x}$. The standard error of $\bar{x}$ is:

$$\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$$

where s is the sample standard deviation. So the **95% confidence interval for a single mean** is:

$$\bar{x} \pm 2\frac{s}{\sqrt{n}}$$

(For small samples, we often use the t distribution so the multiplier is slightly larger than 2; here we use 2 for a simple "by hand" rule.)

Conditions for Formula 2:

1. **Either** the population is (approximately) **normal**, **or** the sample size is $n > 30$ (so the sampling distribution of $\bar{x}$ is approximately normal by the Central Limit Theorem).
2. The sample is a random sample (or can be treated as such) from the population.

Example 1: Confidence interval for a proportion (St. Joe's) — large sample

Goal: Estimate the mortality rate at St. Joe's (Eureka, CA) for this heart procedure using the **large sample** (200 patients, 14 deaths), with **95% confidence**. The national benchmark is $p_0 = 0.08$ (8%).

- **Data:** $n = 200$, $x = 14$ deaths.
- **Statistic:** $\hat{p} = 14/200 = 0.07$ (7%).
- **Standard error:**
$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.07 \times 0.93}{200}} \approx \sqrt{0.0003255} \approx 0.0180.$$
- **95% CI:** $\hat{p} \pm 2 \times \mathrm{SE}(\hat{p}) = 0.07 \pm 2(0.0180) = 0.07 \pm 0.036$.
- **Interval:** Lower bound ≈ 0.034, upper bound ≈ 0.106, so about [**0.034, 0.106**] or [**3.4%, 10.6%**].

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe's is between about 3.4% and 10.6%. The national rate (8%) lies inside this interval, so the data are consistent with St. Joe's having the same mortality rate as the national benchmark.

Small sample for a proportion: Plus Four

When the sample size is small, the condition for the normal approximation may not be satisfied. Then:

- For a **95% confidence interval** for a proportion, use the **Plus Four** formula.
- If the confidence level is not 95% and there are at least 10 successes and 10 failures in the sample, you may use the Wald (z) formula. Otherwise, consult a statistician.

Plus Four formula: Add 2 "successes" and 2 "failures" to the sample. With x = number of successes and n = sample size, define

$$\tilde{p} = \frac{x+2}{n+4}$$

and use the **effective sample size** $n + 4$. The standard error of $\tilde{p}$ is

$$\text{SE}(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

The **95% confidence interval for a proportion (Plus Four)** is:

$$\tilde{p} \pm 2\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}}.$$

Example (small sample): Estimate the mortality rate at St. Joe's using the **small sample** (**16** patients, **3** deaths) with 95% confidence. Use the Plus Four method.

- **Data:** $x = 3$ deaths (successes), $n = 16$ patients.
- **Plus Four estimate:** $\tilde{p} = \frac{x+2}{n+4} = \frac{3+2}{16+4} = \frac{5}{20} = 0.25$.
- **Standard error:** $\text{SE}(\tilde{p}) = \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}} = \sqrt{\frac{0.25\times0.75}{20}} = \sqrt{0.009375} \approx 0.0968$.
- **95% CI:** $\tilde{p} \pm 2 \times \text{SE}(\tilde{p}) = 0.25 \pm 2(0.0968) = 0.25 \pm 0.194$.
- **Interval:** Lower bound ≈ 0.056, upper bound ≈ 0.44, so about [**0.06, 0.44**] or [**6%, 44%**].

Interpretation: We are 95% confident that the true mortality rate for this heart procedure at St. Joe's is between about 6% and 44%. With only 16 patients, the interval is wide and the estimate is uncertain; the Plus Four method gives a plausible range that is more reliable than the Wald interval for such a small sample.

Example 2: Confidence interval for a mean (by hand)

Goal: Estimate the mean recovery time (in days) for patients after the advanced heart procedure using a small sample, with **95% confidence**.

Data (recovery time in days for $n = 5$ patients):
5, 7, 7, 9, 12

By hand:

- **Sample mean:** $\bar{x} = \frac{5+7+7+9+12}{5} = \frac{40}{5} = 8$ days.
- **Sample standard deviation:**
  Deviations from mean: $-3, -1, -1, 1, 4$.
  Squared deviations: 9, 1, 1, 1, 16. Sum = 28.
  $s^2 = 28/(5-1) = 7$, so $s = \sqrt{7} \approx 2.65$ days.
- **Standard error:** $\text{SE}(\bar{x}) = \frac{s}{\sqrt{n}} = \frac{2.65}{\sqrt{5}} \approx 1.18$ days.
- **95% CI:** $\bar{x} \pm 2 \times \text{SE}(\bar{x}) = 8 \pm 2(1.18) = 8 \pm 2.36$ days.
- **Interval:** Lower bound ≈ 5.64 days, upper bound ≈ 10.36 days, so about [**5.6, 10.4**] days.

Interpretation: We are 95% confident that the true mean recovery time (in days) for patients after this advanced heart procedure is between about 5.6 and 10.4 days. The 95% confidence means that if we were to repeat the data collection many times, 95% of the time the true mean recovery time would be within the intervals (e.g. lower and upper bounds) computed from the samples. This interval is based on only 5 patients, so it is wide and uncertain; a larger sample would give a narrower and more precise interval.

Check yourself — Sampling distributions and confidence intervals

1. Define **parameter** vs **statistic** in one sentence each.
2. What is the **sampling distribution** of $\hat{p}$?
3. Give the 95% CI formula for a population **proportion** (Wald form).
4. St. Joe's: $n = 200$, $x = 14$ deaths. Compute $\hat{p}$ and approximate 95% CI.
5. Write one sentence interpreting that CI in context.

Answers — Sampling distributions and confidence intervals

1. **Parameter:** number describing the **population** (usually unknown). **Statistic:** number computed from the **sample** (e.g. $\hat{p}$, $\bar{x}$).
2. The distribution of $\hat{p}$ values you would get from many random samples of the same size from the same population.
3. $\hat{p} \pm 2\sqrt{\hat{p}(1-\hat{p})/n}$ when BINS + large-count conditions hold.
4. $\hat{p} = 0.07$. SE ≈ 0.018. CI $\approx 0.07 \pm 0.036 \rightarrow$ about [**0.034, 0.106**].
5. We are 95% confident the true mortality rate for this procedure at St. Joe's is between about 3.4% and 10.6%.

Section 4: Confidence intervals and the importance of sampling

Confidence intervals and the importance of sampling

> **i** Project 2 connection
>
> Project 2 requires a **simple random sample** from a **listable finite population**. Sampling choices here directly affect whether your CI is meaningful.

In this section

A correct CI formula is not enough — **how you sample** determines whether the interval applies to the population you care about.

What is a confidence interval?

A **confidence interval (CI)** provides a range of plausible values for a population parameter based on a sample from the population. It quantifies the uncertainty in our sample estimate. For a 95% CI, we use the approximate formula:

Statistic ± 2 × (standard error of the statistic).

Whether that interval is actually *reasonable* for the population we care about depends on **how the sample was obtained** — that is, on the **sampling method**.

Motivation: How many business majors?

Suppose we want to estimate the **proportion of all undergraduate students on campus** who are business majors.

We decide to survey the students in **this Stat 109 class**.

- **In this async course:** How many students in the room are majoring in business? (Count; call that x. The class size is n.)
- **Example values:** Suppose there are $n = 25$ students and $x = 8$ report being business majors.

Sample proportion: $\hat{p} = \frac{x}{n} = \frac{8}{25} = 0.32$ (32%).

95% confidence interval for p (Wald, assuming conditions hold):

- $\text{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.32 \times 0.68}{25}} \approx 0.093$.
- **95% CI:** $\hat{p} \pm 2 \times \text{SE}(\hat{p}) = 0.32 \pm 0.186$, so about [**0.13, 0.51**] or [**13%, 51%**].

So from the formula we get: *We are 95% confident that the true proportion of all undergraduates who are business majors is between about 13% and 51%.*

Is this interval reasonable?

Critical question: Is it reasonable to treat this interval as applying to **all students on campus**?

- **Population of interest:** All undergraduate students at the university.
- **Sample:** Only students in this Stat 109 class.

Problem: Business majors are **not required** to take Stat 109. So:

- The students in this class are not a random sample of all undergraduates.
- We have a **convenience sample** (whoever is in the room). The proportion of business majors in Stat 109 might be higher or lower than on the whole campus — we have no guarantee the sample is representative.
- So the **confidence interval from the formula** may be misleading: it describes uncertainty due to *random sampling variation in using* $\hat{p}$ to estimate p, but it does **not** account for **bias** due to how we chose the sample. If the sample is biased, the interval may not be a plausible range for the *campus* proportion at all.

Takeaway: The way we **sample** matters. To make confidence intervals that we can trust for the population we care about, we need to think about **sampling methods** and **bias**.

Sampling frame

The **sampling frame** is the list we use to identify and select units from the population. Ideally, the frame matches the population of interest (every member has a chance to be selected; no one is in the frame who isn't in the population, and vice versa). In the business-major example, "students in this Stat 109 class" is a poor frame for "all undergraduates on campus".

Sampling methods (overview)

We can **classify** how a sample was obtained:

Method	Brief description
Convenience	Select whoever is easy to reach (e.g. this class). Often biased.
Simple random (SRS)	Every subset of n units has the same chance of being selected.
Stratified	Divide population into strata (groups); take a random sample from each stratum.
Cluster	Divide population into clusters; randomly select clusters and often measure all (or a sample of) units in those clusters.
Systematic	From a list, take every kth unit (e.g. every 10th student).
Multi-stage	Combine methods (e.g. randomly select clusters, then randomly select units within clusters).

The next section goes into more detail on these methods and on **carrying out** simple random sampling.

Bias and precision

- **Bias:** Systematic tendency to over- or under-estimate the parameter. Comes from *how* we select or measure (e.g. who is in the frame, who responds).
- **Precision:** How similar our estimates would be if we repeated the process (related to standard error and width of the CI). Affected by sample size and design.

Types of bias (when sampling or surveying):

- **Sampling bias:** The sampling method systematically excludes or over-represents some groups (e.g. convenience sample of one class).

- **Undercoverage:** Some groups in the population are not in the sampling frame or have little chance of being selected.
- **Nonresponse bias:** People who don't respond differ from those who do.
- **Response bias:** The way questions are asked or the setting leads to systematically wrong answers.
- **Voluntary response bias:** Only people who choose to respond are included; they often differ from the population.

Ways to reduce bias: Use a good sampling frame, avoid convenience/voluntary samples when possible, use random sampling (e.g. SRS, stratified), try to reduce nonresponse and careful questionnaire design to reduce response bias.

Ways to increase precision: Increase sample size; sometimes use stratified or other designs that reduce variability.

Finite population correction (FPC)

So far we assumed the sample is a small fraction of the population, so we used:

- **Proportion:** $\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- **Mean:** $\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$

When we **sample without replacement** from a **finite population** of size N, and n is not negligible compared to N, we multiply the standard error by the **finite population correction (FPC)**:

$$\mathrm{FPC} = \sqrt{\frac{N-n}{N-1}}.$$

(When N is large, $\sqrt{\frac{N-n}{N}}$ is a common approximation.)

Interpretation: When we draw a large fraction of the population, we get more information than the "infinite population" formula suggests, so the standard error **shrinks** (the FPC is < 1 when $n > 1$). When n is small relative to N, the FPC is close to 1 and we usually ignore it.

95% confidence interval for a proportion (with FPC):

$$\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \cdot \sqrt{\frac{N-n}{N-1}}.$$

95% confidence interval for a mean (with FPC):

$$\bar{x} \pm 2\frac{s}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}.$$

When to use: When the population is finite and n/N is large (e.g. n is more than about 5–10% of N), the FPC narrows the interval. When n/N is small, the FPC is near 1 and the usual formulas are fine.

Example: Proportion with and without FPC

Setting: A small clinic has $N = 50$ patients who had a certain procedure. We take a random sample of $n = 20$ and find $x = 8$ had a positive outcome. So $\hat{p} = 8/20 = 0.40$.

Without FPC (infinite-population formula):

- $\mathrm{SE}(\hat{p}) = \sqrt{\frac{0.40\times 0.60}{20}} = \sqrt{0.012} \approx 0.1096$.
- Margin $\approx 2(0.1096) = 0.219$. **95% CI:** $0.40 \pm 0.219 \rightarrow$ about [**0.18, 0.62**].

With FPC:

- $\mathrm{FPC} = \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{50-20}{49}} = \sqrt{\frac{30}{49}} \approx 0.782$.

- $\text{SE}(\hat{p}) \times \text{FPC} \approx 0.1096 \times 0.782 \approx 0.0857$. Margin $\approx 2(0.0857) = 0.171$.
- **95% CI (with FPC):** $0.40 \pm 0.171 \to$ about [**0.23, 0.57**].

The FPC narrows the interval because we sampled a large fraction ($20/50 = 40\%$) of the population, so we have more information than the infinite-population formula assumes.

Example: Mean with and without FPC

Setting: A small population of $N = 30$ plots. We randomly select $n = 10$ and measure a variable (e.g. plant count per plot). The data are: **4, 5, 6, 7, 7, 8, 9, 10, 11, 12**.

- **Sample mean:** $\bar{x} = (4+5+6+7+7+8+9+10+11+12)/10 = 79/10 = 7.9$.
- **Sample SD:** Deviations from 7.9: $-3.9, -2.9, -1.9, -0.9, -0.9, 0.1, 1.1, 2.1, 3.1, 4.1$. Squared: 15.21, 8.41, 3.61, 0.81, 0.81, 0.01, 1.21, 4.41, 9.61, 16.81. Sum = 60.9. $s^2 = 60.9/9 \approx 6.77$, so $s \approx 2.60$.
- $\text{SE}(\bar{x}) = s/\sqrt{n} = 2.60/\sqrt{10} \approx 0.822$.

Without FPC (infinite-population formula):

- Margin $\approx 2(0.822) = 1.64$. **95% CI:** $7.9 \pm 1.64 \to$ about [**6.3, 9.5**].

With FPC:

- $\text{FPC} = \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{30-10}{29}} = \sqrt{\frac{20}{29}} \approx 0.830$.
- $\text{SE}(\bar{x}) \times \text{FPC} \approx 0.822 \times 0.830 \approx 0.682$. Margin $\approx 2(0.682) = 1.36$.
- **95% CI (with FPC):** $7.9 \pm 1.36 \to$ about [**6.5, 9.3**].

Again, the FPC gives a slightly narrower interval because we sampled a substantial fraction ($10/30 \approx 33\%$) of the finite population.

Infinite-population formulas (full conditions)

When the population is treated as **infinite** (or n/N is very small), use these versions and conditions:

One proportion (Wald), 95% CI: $\hat{p} \pm 2\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

Conditions:

1. **Binomial random process:** (a) **B**inary outcome, (b) **I**ndependent trials, (c) **N**umber of trials n fixed, (d) **S**ame probability p on each trial.
2. **Normal approximation:** $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$.

One mean, 95% CI: $\bar{x} \pm 2\frac{s}{\sqrt{n}}$ (or use t for small n).

Conditions:

1. **Either** the population is (approximately) **normal**, **or** $n > 30$ (so the sampling distribution of $\bar{x}$ is approximately normal by the CLT).
2. The sample is a random sample from the population (or can be treated as such).

When the population is **finite** and n/N is not small, multiply the margin of error by the **FPC** $\sqrt{(N-n)/(N-1)}$ for both the proportion and the mean.

::: {.callout-tip icon=false} ## Check yourself — Confidence intervals and the importance of sampling
1. Define **sampling frame**.
2. What is **bias** in sampling?
3. Name two ways to **reduce bias** and one way to increase **precision**.
4. Classify: survey everyone who walks into the campus café at noon.
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1. The list (or rule) that identifies every unit that **could** be selected into the sample from the target population.
2. A systematic tendency for the sample statistic to miss the population parameter in a predictable direction because of how the sample was chosen.
3. Reduce bias: use probability sampling (SRS), improve frame coverage. Increase precision: **larger sample size** n.
4. **Convenience sample** — not a probability sample; limited generalizability.

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Answers — Confidence intervals and the importance of sampling
1. The list (or rule) that identifies every unit that **could** be selected into the sample from the target population.
2. A systematic tendency for the sample statistic to miss the population parameter in a predictable direction because of how the sample was chosen.
3. Reduce bias: use probability sampling (SRS), improve frame coverage. Increase precision: **larger sample size** n.
4. **Convenience sample** — not a probability sample; limited generalizability.
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Section 5: Sampling methods in practice

Sampling methods in practice

In this section

Apply the sampling ideas from Section 4 with concrete frames, SRS steps, and design labels.

Sampling frame

The **sampling frame** is the list or procedure from which we actually select units. For a given population (e.g. "all undergraduates," "all patients at a clinic"), we need to ask: *What list or mechanism will we use to choose who gets into the sample?* The frame should match the population as closely as possible. Gaps (undercoverage) or duplicates can cause bias.

Example: To sample undergraduates, the frame might be a registrar's list of all enrolled students (with IDs). "Students in Stat 109" is a poor frame for "all undergraduates."

Simple random sampling (SRS)

In **simple random sampling**, every possible subset of n units from the population (or from the frame) has the **same chance** of being the sample. Equivalently, every unit has the same probability of being selected, and selections are made at random.

How to carry out SRS (by hand or with software):

1. **Get a list** of all units in the sampling frame (numbered 1 to N).
2. **Generate random numbers** in the range 1 to N (e.g. using a random number generator, table, or software). Draw units corresponding to those numbers. If sampling without replacement, skip repeats until you have n distinct units.
3. **Collect data** from the units in the sampling frame that correspond to the n distinct units from step 2.

Example: To select 5 students from a class of 25, number students 1–25. Use a random number generator (or similar) set to "integers from 1 to 25," generate numbers, and collect data from the students with those IDs. If a number repeats, draw again until you have 5 distinct IDs.

Classifying sampling methods

Given a description of how a sample was obtained, **classify** it as one of:

Method	What happens
Convenience	Units chosen because they are easy to access (e.g. "people in this room," "first 50 visitors"). Often not representative.
Simple random (SRS)	Every set of n units from the frame has the same chance of being selected; usually via random numbers.

Method	What happens
Stratified	Population divided into strata (e.g. by major, by year); a random sample is taken from *each* stratum. Improves representation of each group.
Cluster	Population divided into clusters (e.g. schools, wards); we randomly select *clusters* and then observe all (or a random sample of) units in those clusters. Useful when listing individuals is hard but listing clusters is easy.
Systematic	From an ordered list, pick a random start, then take every kth unit (e.g. every 10th name).
Multi-stage	Several stages of random selection (e.g. randomly choose schools, then randomly choose students within those schools).

Examples: Classify, generalizability, and improving the design

For each scenario below: (1) **Classify** the sampling method as **random sampling** (e.g. SRS, stratified, cluster) or **convenience** (or other as appropriate). (2) **Impact on generalizability:** Would results generalize well to the population of interest? Why or why not? (3) **How to reduce bias** and **how to increase precision** (one concrete suggestion for each, if applicable).

Example 1: Campus opinion poll

A student group wants to estimate the proportion of all undergraduates who support building a new recreation center. They stand outside the student union from 11 a.m. to 1 p.m. on two weekdays and ask every 10th person who walks by to complete a short survey. They get 120 responses.

- **Classify:** Is this random sampling or convenience (or something else)? What is the sampling frame?
- **Impact on generalizability:** Who might be over- or under-represented? Can we generalize the sample proportion to *all* undergraduates?
- **Reduce bias:** How could the study be redesigned to reduce bias (e.g. who is included)?
- **Increase precision:** What could be done to increase precision of the estimate?

Example 2: Online course feedback

A university wants to estimate the mean satisfaction score (1–5) for all students who took at least one online course last year. They send an email survey to a **random sample** of 500 students drawn from the registrar's list of all students who were enrolled in at least one online course; 180 students respond.

- **Classify:** Is the *selection* of the 500 random or convenience? What about the 180 who responded?
- **Impact on generalizability:** Even though the initial 500 were a random sample, might the 180 respondents differ from the 320 non-respondents? How does that affect generalizability?
- **Reduce bias:** What type of bias is most relevant here (e.g. nonresponse)? How could bias be reduced?
- **Increase precision:** How could precision be increased (e.g. sample size, design)?

Example 3: Hospital patient satisfaction

A hospital wants to estimate the proportion of patients who are "very satisfied" with their care. They hand out a satisfaction survey to every patient who is discharged from the maternity ward during a 3-month period and receives the survey before leaving; about 70% return the completed survey.

- **Classify:** How were patients selected? Is this random sampling or convenience?
- **Impact on generalizability:** To what population would the sample proportion generalize? To all hospital patients? To all maternity patients?
- **Reduce bias:** What biases might be present (e.g. undercoverage, nonresponse)? Suggest one way to reduce bias.
- **Increase precision:** Suggest one way to increase precision.

Reducing bias and increasing precision (summary)

- **Reduce bias:** Use a frame that covers the population; avoid convenience and voluntary response when possible; use random sampling (SRS, stratified, etc.); minimize nonresponse and design surveys to reduce response bias.
- **Increase precision:** Increase sample size; use designs that reduce variability (e.g. stratified sampling when groups differ a lot).

Summary

- A **confidence interval** is only as good as the **sample**. Bad sampling (e.g. convenience for a campus-wide question) can make the interval misleading even if the math is correct.
- **Sampling frame** = list or procedure used to select units. Should align with the population of interest.
- **SRS** = every subset of size n equally likely; carried out using a list and random numbers.
- **Classify** methods as convenience, SRS, stratified, cluster, systematic, or multi-stage.
- **Bias** (sampling, undercoverage, nonresponse, response, voluntary response) can invalidate inference; **precision** is improved by larger n and sometimes by design (e.g. stratification).

Check yourself — Sampling methods in practice

1. Population: all HS seniors in California. Frame: school roster IDs. Describe one SRS step.
2. Stratified vs cluster sampling — which splits the population into groups first, then samples **whole** groups?
3. Why does a bad frame cause bias even if you use SRS on the frame?

Answers — Sampling methods in practice

1. Number students 1 to N, use random draws without replacement until n IDs are chosen.
2. **Cluster** sampling selects intact clusters; **stratified** samples within each stratum.
3. Some population members are **missing from the frame** and can never be selected.

ℹ Optional enrichment (not required for Project 2)

- Type I and Type II errors — hypothesis-testing vocabulary
- Rubber-band CI analogy — intuition for confidence intervals